

Lily Sharma

Embedded Systems & IoT Firmware Engineer • 3 Years Experience • MERIQ Index: 97% (Mastered)

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TECHNICAL PROFICIENCIES & CERTIFICATIONS

Core Technologies: ARM Cortex, C/C++, FreeRTOS, Embedded Linux, MQTT, I2C/SPI/UART, PCB Debugging

Verified Certification: Arm Accredited Engineer (AAE)

PROFESSIONAL WORK EXPERIENCE

Embedded Systems & IoT Firmware Engineer — *Texas Instruments* (2023 - Present (3 yrs))

- Spearheaded architectural redesign of core ARM Cortex service, reducing latency by 25%.
- Led a squad of 3 engineers delivering scalable microservices processing 5M requests/day.
- Implemented automated testing & CI/CD pipelines increasing deployment frequency by 34%.

Associate Embedded Systems & IoT Firmware Engineer — *STMicroelectronics* (2023 - 2023)

- Designed and maintained C/C++ data endpoints with 97% SLA uptime.
- Optimized database indexing and caching layers using FreeRTOS saving \$29k/yr cloud costs.

NOTABLE TECHNICAL PROJECTS

Low-Power BLE Sensor Node Firmware for Industrial Telemetry in FreeRTOS [ARM Cortex, C/C++, FreeRTOS, Embedded Linux]
Engineered end-to-end production solution leveraging ARM Cortex and C/C++ with full test automation.

Smart Grid Power Monitoring Device with MQTT & Hardware Cryptographic Accelerator [FreeRTOS, Embedded Linux, MQTT, I2C/SPI/UART]
Architected high-resilience system integrating FreeRTOS with 99.9% uptime.

EDUCATION & CREDENTIALS

B.Tech in Computer Science and Engineering

Purdue University — 2019 - 2023 (Score: 9.7 / 10.0)